

6.3 Analysis

This section develops and complements the spectroscopic areas of organic chemistry previously encountered (see Module 4: Core organic chemistry; 4.2.4 Analytical techniques).

This section demonstrates how analytical techniques introduced in Module 4 (infrared spectroscopy, mass spectrometry and elemental analysis) may be used in combination with NMR spectroscopy to provide evidence of structural features in molecules.

The instrumentation methods of analysis studied during the A level course provide learners with an important base of knowledge, understanding and awareness for further study in Higher Education and in many areas of employment in the broad scientific field.

This section also looks at how unknown organic functional groups can be analysed and identified using simple test-tube tests.

6.3.1 Chromatography and qualitative analysis

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
Types of chromatography	
(a) interpretation of one-way TLC chromatograms in terms of R_f values	M3.1 PAG6 HSW3 Interpretation of TLC to analyse organic compounds.
(b) interpretation of gas chromatograms in terms of: (i) retention times (ii) the amounts and proportions of the components in a mixture.	M3.1, M3.2 To include creation and use of external calibration curves to confirm concentrations of components. Peak integration values will be supplied. HSW3 Interpretation of GC to analyse organic compounds.
Tests for organic functional groups	
(c) qualitative analysis of organic functional groups on a test-tube scale; processes and techniques needed to identify the following functional groups in an unknown compound: (i) alkenes by reaction with bromine (see also 4.1.3 f) (ii) haloalkanes by reaction with aqueous silver nitrate in ethanol (see also 4.2.2 a) (iii) phenols by weak acidity but no reaction with CO_3^{2-} (see also 6.1.1 h) (iv) carbonyl compounds by reaction with 2,4-DNP (see also 6.1.2 d) (v) aldehydes by reaction with Tollens' reagent (see also 6.1.2 e)	PAG7 HSW4 Qualitative analysis.

- (vi) primary and secondary alcohols and aldehydes by reaction with acidified dichromate **(see also 4.2.1 c, 6.1.2a)**
- (vii) carboxylic acids by reaction with CO_3^{2-} **(see also 6.1.3 b).**